Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Three-String Interaction

Proseminar String Theory

Hrvoje Krizic

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Advisors: Prof. Dr. M. Gaberdiel, Dr. M. Lewandowski
Department of Physics, ETH Zürich

Abstract

In this paper, we explain how the world-sheet of a string can be viewed as a Riemann surface. This is essential when calculating quantum mechanical scattering amplitudes, which in string theory characterize the probability of different scattering occurring. In order to map the upper half-plane $\bar{\mathbb{H}}$ of the complex plane onto a polygon, the Schwarz-Christoffel map is introduced. As a result, we can find a canonical representation of the world-sheet of three strings interacting by considering it as a degenerate polygon.

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Chapter 1

Introduction

Interactions and their mediating forces are what make the world a dynamic place. From the atomic interactions that give rise to the properties of matter to the gravitational interactions that govern the movements of celestial bodies, understanding these interactions is essential for comprehending the complex interplay of physical phenomena that surrounds us. It is therefore essential to also deal with interactions in string theory. String theory provides an elegant framework for understanding interactions, as they are described solely by the joining and splitting of strings. We will see, that such interactions can be described in a single world sheet that combines the world sheet of all the interacting strings. One example of such a world sheet is shown in Figure 1.1.

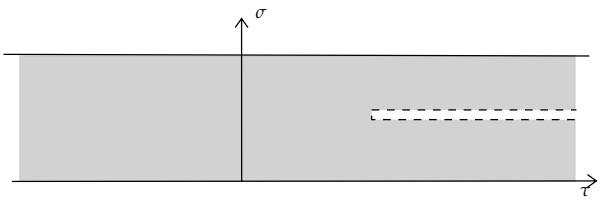


Figure 1.1: The world-sheet of a string interaction is shown. A string splits into two strings.

One of the key insights of string theory is that the dynamics of a string can be fully described by the properties of its world-sheet. This is where the idea of treating world-sheets as Riemann surfaces comes in. Riemann surfaces are two-dimensional real analytic manifolds, which contain a complex structure. By treating world-sheets as Riemann surfaces, one can use Riemann Analysis to understand the interactions of strings. This is very important for calculating quantum mechanical scattering amplitudes, which describe the probability of different scattering events occurring in string theory.

Chapter 2

Particle interactions and observables

A main goal when studying interactions in physics is to convert a collection of interacting particles (in this case, strings) into a probability that an event will occur. Three steps are required to do this:

1. Drawing a diagram of the interaction and figuring out the conformal map that gives a canonical representation of that diagram.
2. Calculating the scattering amplitude from the canonical representation obtained from the first by using conformal field theory.
3. Converting the scattering amplitude into an observable.

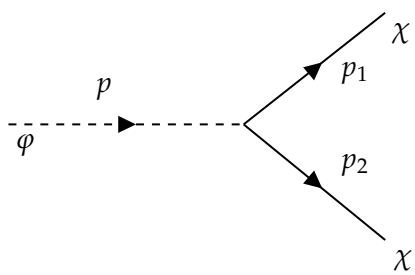
In this paper, primarily the first step is discussed for string interactions.

2.1 Decay of a particle

Before examining strings, we may look at point particles in four dimensions first. Consider as an example a theory, in which two different particles exist, χ and φ. φ are particles with mass M and χ particles with mass m, such that $M \gg 2m$. This ensures that a decay

$$\varphi \to \chi + \chi \tag{2.1}$$

can occur. This decay can be visualized with the corresponding Feynman diagram:



The dashed line represents the φ-particle with momentum p. The two outgoing lines are two χ-particles with momenta p_1 and p_2. To calculate the scattering amplitude of this decay, we introduce a constant λ in our theory. This constant quantifies the strength of the interaction. According to QFT, the scattering amplitude, or in other words, the amplitude for an initial state consisting of a φ-particle to become a final state consisting of the two outgoing χ-particles, is

$$_{\text{out}}\langle p_1, p_2 | p \rangle_{\text{in}} = -i(2\pi)^4 \delta(p - p_1 - p_2) T(p, p_1, p_2), \tag{2.2}$$

where T is exactly the constant λ. λ is our constant in the theory of two outgoing particles. The factor $\delta(p - p_1 - p_2)$ ensures the momentum and energy conservation, since p_i are four-momenta. In the third step, the goal is to turn the amplitude into an observable. The simplest observable in this case is the lifetime of the φ-particle, which (without proof) is given by

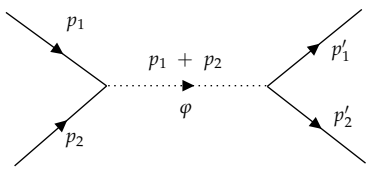
$$\frac{1}{\tau_\varphi} = \frac{\lambda^2}{32\pi} \frac{1}{M} \sqrt{1 - \frac{4m^2}{M^2}}. \tag{2.3}$$

This result is reasonable, since for $\lambda \to 0$ we have the lifetime τ_φ becoming infinite.

2.2 Two particles scattering

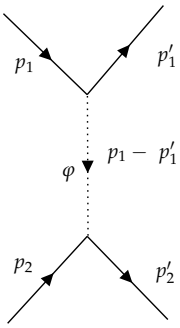
A more interesting case is the scattering of two χ-particles of each other. The initial and final state consists of two χ-particles with momenta p_1 and p_2, respectively, after the scattering p_1' and p_2'. As the φ-particle acts as the intermediate particle, the scattering process will use the $\varphi\chi\chi$ interaction twice. This scattering process can be represented with three different Feynman-Diagrams:

1. The simplest Feynman diagram, that represents this scattering is



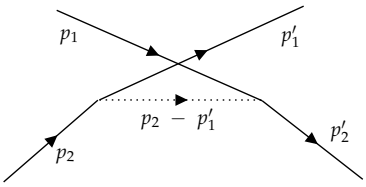
   All the in- and outgoing χ particles are on the mass-shell, which means that they satisfy $p^2 = -m^2$ and thus represent physical states. The intermediate particle φ typically is not on the mass-shell, since $(p_1 + p_2)^2 \neq -M^2$ in general unless the initial state is carefully prepared, such that the equation $(p_1 + p_2)^2 = -M^2$ holds.

2. The following Feynman diagram shows a second way to visualize the scattering:



One χ-particle turns into an outgoing χ-particle with momentum p_1' and a φ-particle with momentum $p_1 - p_1'$. This process contradicts the energy conservation rules in the center of mass system. A lighter particle cannot decay into one particle with the same mass and one heavier particle, the φ-particle. However, we can treat the φ-particle as a virtual particle which joins the incoming second χ-particle with momentum p_2 to give the outgoing χ-particle with momentum p_2'.

3. The last diagram



shows a third possibility of this scattering. A χ-particle with momentum p_2 creates an outgoing χ-particle with momentum p_1' and an intermediate φ-particle with momentum $p_2 - p_1'$. This intermediate particle joins the other χ-particle with momentum p_1 and creates a χ-particle with momentum p_2'.

With these three diagrams, step 1 is completed. Moving on to the second step, we can calculate the scattering amplitude using the Feynman rules of field theory, which give us

$$_{\text{out}}\langle p_1, p_2 | p \rangle_{\text{in}} = -i(2\pi)^4 \delta(p_1 + p_2 - p_1' - p_2') T(p_1, p_2; p_1', p_2'). \tag{2.4}$$

By defining the variables

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p_1')^2, \quad u = -(p_2 - p_1')^2 \tag{2.5}$$

we get

$$T(p_1, p_2; p_1', p_2') = \lambda^2 \left(\frac{1}{s - M^2} + \frac{1}{t - M^2} + \frac{1}{u - M^2} \right) \tag{2.6}$$

This is a rather nice result because it allows us to calculate, for instance, the intermediate φ-particle's mass M simply by knowing the mass of the χ-particles and the scattering amplitude. Nevertheless, one cannot observe the

scattering amplitude. In a third step, we have to convert this scattering amplitude to a scattering cross-section which is observable. This cross-section can be calculated using field theory methods:

$$\frac{d\sigma}{d\Omega} = \frac{1}{4s}\left|\frac{T}{4\pi}\right|^2 \tag{2.7}$$

with $s = -(p_1 + p_2)^2$ as defined. It is worth mentioning, that in the center-of-mass frame $s = -(p_1 + p_2)^2 = (E_1 + E_2)^2$. The Lorentz invariant s is therefore the square of the total energy available in the center-of-mass frame. Equation 2.7 completes the third step.

Chapter 3

String interactions and global world-sheets

String interactions involve splitting and joining of strings. As we have seen in the introduction, we can draw string diagrams by taking all the world-sheets and putting them together into one world sheet. Since gauge choices affect the world-sheets, one has to verify that our interaction pictures are invariant under the gauge choice. We begin by using the static gauge

$$X^0(\tau, \sigma) = c\tau. \tag{3.1}$$

In the static gauge, the parameterization of the strings depends on the energy. We have

$$\sigma \in [0, 2\pi\alpha' E], \tag{3.2}$$

where α' is the slope parameter $\alpha' \propto \frac{1}{T_0}$ with T_0 being the string tension. What are the takeaways from this? Since we have energy conservation, the σ length of the total string has to be conserved. As an example we can take a string with energy E splitting into two strings of energies E_1 and E_2, then $E = E_1 + E_2$ and with this

$$\sigma^{\text{before}}_{\text{max.,tot.}} = 2\pi\alpha' E = 2\pi\alpha' E_1 + 2\pi\alpha' E_2 = \sigma^{\text{after}}_{\text{max.,tot.}}$$

This interaction is shown in Figure 3.1.

Let us now consider the light-cone gauge for open strings $\beta = 2$:

$$X^+ = 2\alpha' p^+ \tau \tag{3.3}$$

p^+ is defined as

$$p^+ = \pi \mathcal{P}^{\tau +} \tag{3.4}$$

which implies $\sigma \in [0, \pi]$. But we have seen that in the static gauge, the total σ length is conserved, which is a contradiction to the light-cone gauge. Furthermore, there is another issue with the light-cone gauge for open strings.

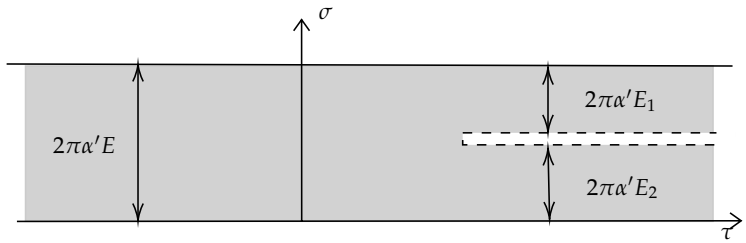


Figure 3.1: Splitting of a string with energy E into two strings with energies E_1 and E_2.

If we have a three-string interactions, the physical time X^+ must be the same for all string at τ_q. Hence,

$$\begin{aligned} X_1^+ &= 2\alpha' p_1^+ \tau_q \\ \stackrel{!}{=} X_2^+ &= 2\alpha' p_2^+ \tau_q \\ \stackrel{!}{=} X_3^+ &= 2\alpha' p_3^+ \tau_q. \end{aligned} \tag{3.5}$$

Given that we combined all strings into a single world-sheet, τ_q is the same for all strings. But this leads to the conclusion that $p_1^+ = p_2^+ = p_3^+$, which is generally wrong. This means that since we seek invariance when choosing gauges, our light-cone gauge is incorrect. How can we change our light-cone setup to handle interactions? We recall that the general light-cone gauge (for open and closed strings) is given by

$$X^+ = \beta\alpha' p^+ \tau, \quad p^+ = \frac{2\pi}{\beta}\mathcal{P}^+ \tag{3.6}$$

For open strings, we chose $\beta = 2$ in order to get $\sigma \in [0, \pi]$. But this was just a convention. We can modify β, such that σ is constant in total and X^+ is not dependent on p^+ anymore, which resolves the second problem mentioned above. By choosing

$$\beta = \frac{1}{\alpha' p^+} \tag{3.7}$$

we get

$$\sigma \in [0, 2\pi\alpha' p^+] \tag{3.8}$$

which is good since p^+ is conserved in an interaction and therefore the total σ-length is conserved. In Figure 3.2 an example of such a world sheet is shown, where two strings form a single open string and this intermediate string then splits into two open strings.

Furthermore, we have

$$\begin{aligned} X_1^+ &= \tau \\ X_2^+ &= \tau \\ X_3^+ &= \tau. \end{aligned} \tag{3.9}$$

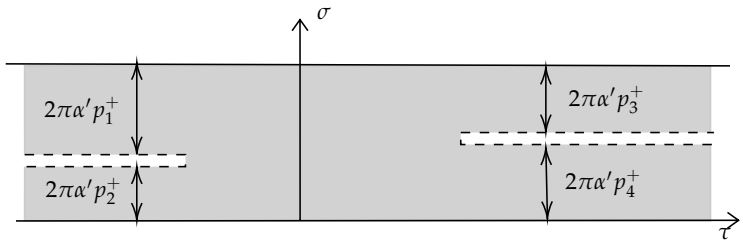


Figure 3.2: An intermediate open string is formed when incoming open strings with momenta p_1^+ and p_2^+ combine. This intermediate open string divides into two strings with momenta p_3^+ and p_4^+.

At $\tau = \tau_q$ (where the interaction occurs), all three X^+ times are the same, which is what we needed.

By defining $\beta = 1/\alpha' p^+$ and combining many world-sheets into a single world-sheet, we have found a method for drawing string diagrams. Nevertheless, we have not yet found a canonical representation of the string diagram. This is going to be discussed in the next sections.

Chapter 4

World-sheets as Riemann surfaces

In the next two chapters, we will show that the world-sheet can be treated as Riemann surfaces. This will make it easier for us to deal with interactions and to find a way to describe a world-sheet canonically.

4.1 Riemann surfaces

Before we define what a Riemann surface actually is, we need some definitions from complex analysis.

Definition 4.1 *Let $\Omega \in \mathbb{C}$ be open. An **analytic function** is a function $f : \Omega \to \mathbb{C}$ of the complex variable z, which has the property, that for all z_0 the complex derivative $f'(z)$ exists in some neighborhood of z_0.*

These analytic functions satisfy the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \tag{4.1}$$

where $f(z)$ is of the form $f(x+iy) = u(x,y) + iv(x,y)$. Riemann surfaces are the simplest geometric objects that locally possess the structure of complex numbers. To obtain such a structure, we need so-called complex charts [1].

Definition 4.2 *A **complex chart** on a topological space X is a homeomorphism $\varphi : U \to V$ where U is an open set in X and V is an open set in $\mathbb{C}$.*

Definition 4.3 *A **Riemann surface** is a two-dimensional real manifold, which is equipped with charts $\varphi_\alpha : U_\alpha \to \mathcal{U} \in \mathbb{C}$, such that the open sets U_α cover the Riemann surface and that over the intersections $U_\alpha \cap U_\beta$ the transition functions $\varphi_\alpha \circ \varphi_\beta^{-1}$ are analytic.*

We say that two Riemann surfaces are equivalent if there is a continuous one-to-one analytic mapping relating them. Analytic maps are conformal,

which means that angles between arcs are conserved (at least locally) except at possible points, where the derivative of the map vanishes. What happens with distances under conformal mapping? Let $f(z)$ denote such a conformal map. Then for $w = f(z)$ we have

$$|dw| = |f'(z_0)||dz|. \tag{4.2}$$

Therefore, $|dw|$, which is the length of the image of dz, is scaled by a factor of $|f'(z_0)|$. Thus, lengths are not conserved under conformal maps. Weyl transformations

$$h_{\alpha\beta} \to \Omega^2 h_{\alpha\beta} \tag{4.3}$$

have the same property. They scale all metric components by the same factor at each point on the world-sheet, while leaving the string action (Polyakov action) invariant. Furthermore, from this invariance of the string action under the local scale transformation, the scale of lengths on the world-sheet is not physical. This shows that it is logical to interpret world-sheets as Riemann surfaces because the scale of lengths is not well-defined on Riemann surfaces either. Nevertheless, there is one difference between Riemann surfaces and world-sheets: the metric. For world-sheets, on the one hand, we use the Minkowski metric

$$-ds^2 = -d\tau^2 + d\sigma^2, \tag{4.4}$$

but for Riemann surfaces, we use the Euclidean metric,

$$ds^2 = dx^2 + dy^2. \tag{4.5}$$

Because of this, the claim that world-sheets may be thought of as Riemann surfaces cannot be fully proven. However, nothing prevents us from attempting to treat them as such, particularly given the fact that it is well known that much about Minkowski theory can be learned from the Euclidean equivalent. As it turns out, describing world-sheets as Riemann surfaces yields a consistent picture of string interactions.

4.2 Mapping open strings to $\bar{\mathbb{H}}$

The simplest way to assign complex coordinates w to our world-sheet is

$$w = \tau + i\sigma, \tag{4.6}$$

where the open string is an infinite strip $\mathbb{W}$ with imaginary part

$$\mathrm{Im}(w) \in [0, 2\pi\alpha' p^+]. \tag{4.7}$$

It would be helpful to first conformally map this strip to the entire complex plane or, as we will see, to the upper half-plane $\bar{\mathbb{H}}$ in order to obtain a

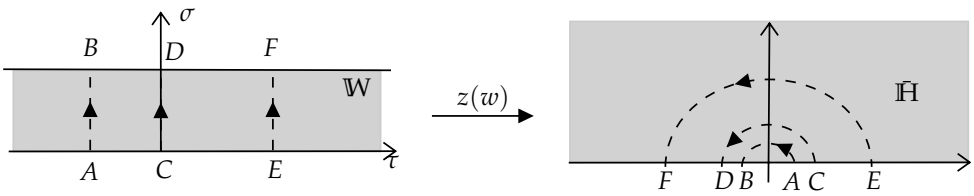


Figure 4.1: The infinite strip is mapped into $\bar{\mathbb{H}}$ via the exponential map 4.8.

canonical representation of string interactions. An example of a map that conformally maps $\mathbb{W}$ to $\bar{\mathbb{H}}$ is given by

$$z(w) = \exp\left(\frac{w}{2\alpha' p^+}\right) = \exp\left(\frac{\tau}{2\alpha' p^+}\right)\exp\left(i\frac{\sigma}{2\alpha' p^+}\right). \tag{4.8}$$

We see that the first term is the radial component of z. At $\tau \to -\infty$ the radial part vanishes and we have $z \to 0$. The $\theta = \sigma/(2\alpha' p^+)$-component is the angle made with the real axis. Since $\sigma \in [0, 2\pi\alpha' p^+]$ we have $\theta \in [0, \pi]$ which proves, that we map onto $\bar{\mathbb{H}}$. The two boundaries of the strip are mapped to the real line. The strips are now semicircles, as shown in Figure 4.1. The benefit of $\bar{\mathbb{H}}$ is that it is much simpler to use. It is much easier to work with conformal maps from the upper half-plane to other Riemann surfaces than with the world-sheet $\mathbb{W}$. For example, we can map the upper half-plane onto the unit disk $\mathbb{D}$ via the transformation

$$\eta(z) = \frac{1+iz}{1-iz}. \tag{4.9}$$

The real line (the boundaries of the world-sheet representation) is mapped onto the unit circle since for $z = x \in \mathbb{R}$ we have

$$|\eta|^2 = \eta\eta^* = \frac{1+ix}{1-ix}\cdot\frac{1+ix}{1-ix} = 1 \tag{4.10}$$

$\eta = 1$ corresponds to the infinite past, and $\eta = -1$ to the infinite future. It is worth mentioning, that the infinite past and infinite future are mapped onto two points on the boundary of the bounded region. Since these limits cannot be reached for any finite time, they have to be removed. We call such points *punctures*.

4.2.1 Closed strings

The conformal map

$$\tilde{z}(w) = \exp\left(\frac{w}{\alpha' p^+}\right) = \exp\left(\frac{\tau}{\alpha' p^+}\right)\exp\left(i\frac{\sigma}{\alpha' p^+}\right) \tag{4.11}$$

maps a closed string to the full complex z-plane, since the angle to the real axis is now $\theta \in \sigma/\alpha' p^+$ and with $\sigma \in [0, 2\pi\alpha' p^+]$ we have $\theta \in [0, 2\pi]$ so

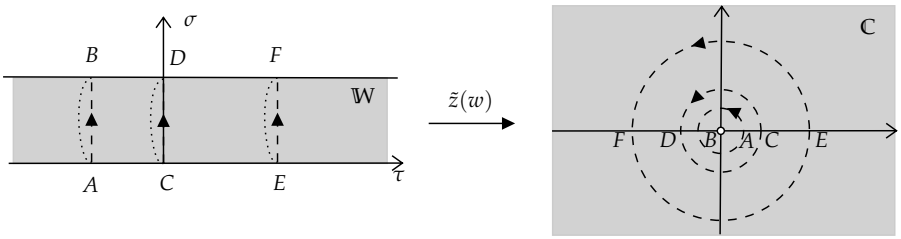


Figure 4.2: $\mathbb{W}$ is mapped into $\mathbb{C}$ via the map $\tilde{z}(w)$ in equation 4.11. The origin has to be removed in $\mathbb{C}$ since it is a puncture. Furthermore, the dotted lines in $\mathbb{W}$ indicate, that the string is closed, so we identify points on the two edges that have the same value of τ.

closed strings are circles. Furthermore, the radial part still grows with τ and therefore, $\tilde{z}(w)$ maps $\mathbb{W}$ to the complex plane with the origin removed since it is a still a puncture ($\tau \to -\infty$). This mapping is shown in Figure 4.2.

Chapter 5

Schwarz-Christoffel map and three-string interaction

We are now ready to examine conformal maps for world-sheets of interacting strings. The aim will be to find a conformal map that will map Figure 1.1 onto the upper half-plane $\bar{\mathbb{H}}$.

5.1 Schwarz-Christoffel map

Consider a polygon in a complex $\mathbb{W}$-plane. Let the polygon also have n sides and n vertices, which are denoted by $\{P_i\}_{i=1}^n$. The conformal map mapping $\bar{\mathbb{H}}$ to this polygon is given by the Schwarz-Christoffel map, which is illustrated in Figure 5.1. The real line gets mapped onto the polygon boundary. The vertices $\{P_i\}_{i=1}^n$ of the polygon have turning angles $\{\alpha_i\}_{i=1}^n$. We restrict them by definition to

$$-\pi \leq \alpha \leq \pi. \tag{5.1}$$

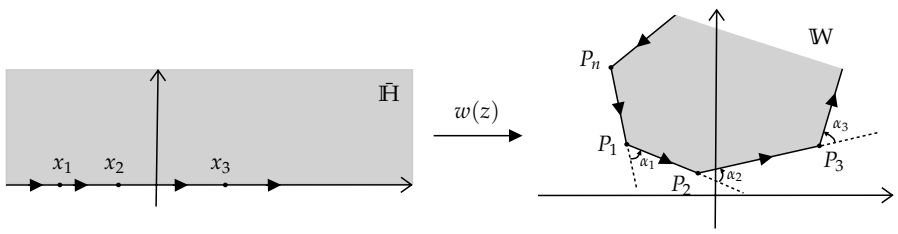


Figure 5.1: The upper half-plane is mapped onto the polygon in the $\mathbb{W}$-plane by using the Schwarz-Christoffel map. The $\{x_i\}_{i=1}^n$ are well-ordered. The turning angles $\{\alpha_i\}_{i=1}^n$ at the vertices $\{P_i\}_{i=1}^n$ are restricted to $-\pi \leq \alpha \leq \pi$.

The Schwarz-Christoffel map is then given by the differential equation

$$\frac{dw}{dz} = A(z - x_1)^{\frac{-\alpha_1}{\pi}}(z - x_2)^{\frac{-\alpha_2}{\pi}} \dots (z - x_n)^{\frac{-\alpha_2}{\pi}}, \tag{5.2}$$

where $A \in \mathbb{R}$, $x_i = z(P_i)$, $i \in 1, 2, ..., n-1$ and, very importantly, well-ordered:

$$x_1 < x_2 < ... < x_{n-1}. \tag{5.3}$$

Furthermore, we define $z(P_n) \hat{=} \infty$. However, it is important to mention here that α_n is defined by the constraint

$$\alpha_n = 2\pi - \alpha_1 - \dots - \alpha_{n-1} \tag{5.4}$$

Why is the solution of the differential equation 5.2 the right mapping of $\bar{\mathbb{H}}$ to the polygon in $\mathbb{W}$? We first limit our discussion to the real axis in the upper half-plane that is mapped onto the polygon's edges. Since $dz \in \mathbb{R}$ and therefore $\arg dz = 0$, we get

$$\operatorname{Arg} dw = \operatorname{Arg}(A) - \frac{\alpha_1}{\pi}\operatorname{Arg}(z - x_1) - \dots - \frac{\alpha_{n-1}}{\pi}\operatorname{Arg} z - x_{n-1} \tag{5.5}$$

by taking the main branch of the argument on both sides of equation 5.2 and by using the properties

$$\operatorname{Arg}(z_1 \cdot z_2) = \operatorname{Arg}(z_1) + \operatorname{Arg}(z_2) \tag{5.6}$$

and

$$\operatorname{Arg}(z^\beta) = \beta \operatorname{Arg}(z), \quad \text{for } |\beta| \leq 1. \tag{5.7}$$

The left side of equation 5.5 is $\operatorname{Arg} dw$, which is the angle that an edge of the polygon makes with respect to the horizontal axis of $\mathbb{W}$. On the right side of equation 5.5 we notice that

$$\operatorname{Arg}(x - x_i) = \begin{cases} 0 & x \geq x_i \\ \pi & x < x_i \end{cases}. \tag{5.8}$$

As long as x varies without crossing a turning point x_i nothing changes on the right side. This is important because we do not change the angle to the horizontal axis at the edges of our polygon. What happens if we cross an x_i? $\operatorname{Arg}(x - x_i)$ changes abruptly from π to 0. The right side increases by

$$-\frac{\alpha_i}{\pi}(0 - \pi) = +\alpha_i, \tag{5.9}$$

exactly what we wanted. This shows, that the Schwarz-Christoffel map is reasonable.

5.2 Three-string interactions

How can we apply the Schwarz-Christoffel transformation to the case of an open string splitting into two strings? The idea is to view the world sheet of this interaction as a degenerate polygon with four vertices, shown in Figure 5.2. By taking

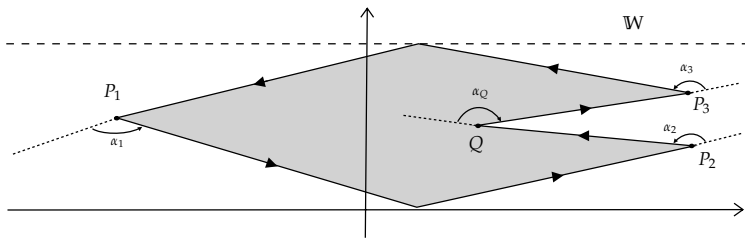


Figure 5.2: A polygon with four vertices is shown. In the degenerate case, where $\alpha_1 = +\pi, \alpha_2 = +\pi, \alpha_3 = +\pi, \alpha_Q = -\pi$ we get the world-sheet of three interacting strings (one splitting into two).

$$\alpha_1 = +\pi, \quad \alpha_2 = +\pi, \quad \alpha_3 = +\pi, \quad \alpha_Q = -\pi, \tag{5.10}$$

we get the world-sheet of the three interacting strings. To map this polygon to $\bar{\mathbb{H}}$ we apply the Schwarz-Christoffel differential equation for this case. We choose[1]

$$x_2 = z(P_2) = -1, \quad x_3 = z(P_3) = 1, \quad x_1 = z(P_1) \hat{=} \infty. \tag{5.11}$$

x^*, the coordinate of the last vertex, however, cannot be chosen freely. Because of the ordering of x_i we need

$$-1 < x^* < +1. \tag{5.12}$$

The $\bar{\mathbb{H}}$-plane is shown in Figure 5.3.

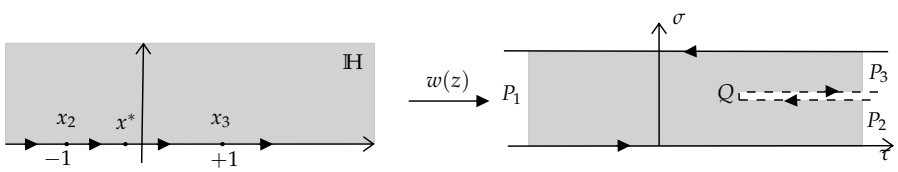


Figure 5.3: The Schwarz-Christoffel map applied to a three string interaction is shown. At the vertices $P_i = z(x_i)$ we choose the angles α_i to be degenerate, so $\alpha_i = +\pi$. $x^* = z(Q)$ is the interaction point. The corresponding degenerate turning angle is $\alpha_Q = -\pi$.

[1]This freedom of choice is not trivial. Nevertheless, it follows from a theorem: three points on the real line can be mapped to any specific coordinates while preserving $\bar{\mathbb{H}}$ and the ordering of the points x_i (See [2])

Substituting the angles 5.10 into equation 5.2 we obtain

$$\frac{dw}{dz} = \frac{A(z-x^*)}{(z+1)(z-1)} \tag{5.13}$$

Decomposing the right side partially yields

$$\frac{dw}{dz} = \frac{A}{2}\left(\frac{(1+x^*)}{(z+1)} + \frac{(1-x^*)}{(z-1)}\right). \tag{5.14}$$

By integrating the right side over dz we obtain

$$w(z) = \frac{A}{2}((1+x^*)\log(z+1) + (1-x^*)\log(z-1)) \tag{5.15}$$

where we use the main branch of the logarithm

$$\log(z) = \log(|z|) + i\,\mathrm{Arg}(z). \tag{5.16}$$

A and x^* in equation 5.15 can be calculated using the fact, that $\sigma := \mathrm{Im}(w)$ has to jump from 0 to $2\pi\alpha' p_2^+$ while crossing $z = x_2 = -1$ on $\bar{\mathbb{H}}$. Applying equation 5.16, we get

$$\sigma = \mathrm{Im}(w) = \frac{A}{2}((1+x^*)\,\mathrm{Arg}(z+1) + (1-x^*)\,\mathrm{Arg}(z-1)) \tag{5.17}$$

At $z = -1$, $\mathrm{Arg}(z+1)$ jumps from π to 0 hence

$$\begin{aligned} \Delta\sigma &= \frac{A}{2}(1+x^*)(0-\pi) \overset{!}{=} 2\pi\alpha' p_2^+ \\ \implies \frac{A}{2}(1+x^*) &= -2\alpha' p_2^+ \end{aligned} \tag{5.18}$$

Similarly at $z = 1$, $\mathrm{Arg}(z-1)$ jumps from π to 0, thus

$$\begin{aligned} \Delta\sigma &= \frac{A}{2}(1-x^*)(0-\pi) \overset{!}{=} 2\pi\alpha' p_3^+ \\ \implies \frac{A}{2}(1-x^*) &= -2\alpha' p_3^+, \end{aligned} \tag{5.19}$$

which leads to the final result

$$w(z) = -2\alpha' p_2^+ \ln(z+1) - 2\alpha' p_3^+ \ln(z-1). \tag{5.20}$$

This can be generalized for n strings:

$$w(z) = \sum_{i=2}^{n} -2\alpha' p_i^+ \ln(z - x_i). \tag{5.21}$$

x^* can be easily obtained by taking the ratio of equations 5.18 and 5.19

$$\frac{-2\alpha' p_2^+}{-2\alpha' p_3^+} = \frac{1+x^*}{1-x^*} \implies x^* = \frac{p_2^+ - p_3^+}{p_2^+ + p_3^+}. \tag{5.22}$$

To check this, we can insert $p_2^+ = p_3^+$ into the equation, which implies $x^* = 0$ and therefore equation 5.22 makes sense.

Bibliography

[1] Rick Miranda. *Algebraic Curves and Riemann Surfaces*. American Mathematical Society, 1995.

[2] Barton Zwiebach. *A First Course in String Theory*. Cambridge University Press, 2004.